UNLOX

LOGISTIC REGRESSION PROJECT REPORT

Project: Diabetes Prediction using Kaggle Data
Student: Pranathi R
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Objective	Dataset	Model
Predict Outcome: 0 = No diabetes, 1 = Diabetes.	Pima Indians Diabetes Database from Kaggle; file: diabetes.csv.	Median imputation + StandardScaler + balanced Logistic Regression.

Dataset and Workflow

Inputs: Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, and Age.

Workflow: load Kaggle CSV → inspect data → replace invalid zeros → train-test split → median imputation → scaling → train Logistic Regression → evaluate → interpret coefficients.

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Evaluation Metrics

Metric	Result	Meaning
Accuracy	0.734	Overall correct predictions
Precision	0.603	Reliability of positive predictions
Recall	0.704	Actual positive cases detected
F1-score	0.650	Balance of precision and recall
ROC-AUC	0.813	Overall class-separation ability

Confusion Matrix

	Predicted 0	Predicted 1
Actual 0	TN = 75	FP = 25
Actual 1	FN = 16	TP = 38

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Key Findings

- The model obtains a ROC-AUC of about 0.813 and detects approximately 70.4% of the positive cases in the fixed test split.
- Glucose and BMI have the strongest positive model coefficients. Recall is especially important in a screening-style classification task because false negatives represent missed positive cases.

Limitation: This small historical dataset represents a specific population. The model is suitable for learning classification concepts, not for medical diagnosis.

Submission Checklist

- Submit the executed .ipynb file and this one-page PDF.
- Add your details.
- Keep all metric outputs and graphs visible.
- Write the final conclusion in your own words.

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